RIDWAN AHMED

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PROFILE

Mechanical engineer currently in the M2 (final year) of Structural Mechanics and Coupled Systems at Cnam Paris (LMSSC laboratory), with hands-on experience in FEM simulation (ANSYS, Patran/Nastran), CAD design (SolidWorks, AutoCAD), and numerical/data analysis (Python, MATLAB). Published first-author researcher in composite materials fabrication and characterization (stir casting, SEM, UTM testing), with cross-functional industry experience spanning operations, project management, and quality assurance. Seeking a research-oriented internship applying simulation and structural analysis skills to real engineering problems.

PUBLICATIONS

- Ahmed, R. et al. "Effect of jute stick charcoal powder on aluminum metal matrix composite characterization." Results in Engineering, Elsevier, 2024. (First author)

EDUCATION

MS, Structural Mechanics and Coupled Systems — Currently in M2 (Final Year) *Sep 2025 – Sep 2027 (Expected)*
Conservatoire National des Arts et Métiers (Cnam), Paris, France

B.Sc. (Engineering), Mechanical Engineering *2019 – 2023*
Bangladesh Army University of Science and Technology

TECHNICAL SKILLS

FEM / Simulation: ANSYS, MSC Patran/Nastran (structural & vibration simulation, notch/stress analysis on turbine blades)
CAD: SolidWorks, AutoCAD
Programming / Analysis: Python, MATLAB, Excel
Materials & Manufacturing: Composite materials fabrication & characterization, stir casting, mold & pattern preparation, additive manufacturing
Experimental / Lab: Universal Testing Machine (UTM) data analysis, Scanning Electron Microscope (SEM) imaging, wind tunnel testing
Languages: English (IELTS Academic 8.0), Bangla (Native), French (A2)

RESEARCH & PROJECTS

Notch Effect Simulation on Wind Turbine Blades (HAWT & VAWT)

- Simulated the structural effects of notches on horizontal- and vertical-axis wind turbine blades using ANSYS to evaluate stress concentration and failure behavior.

Jute Stick Charcoal-Reinforced Aluminum Matrix Composites — Undergraduate Thesis

- Designed and ran a stir-casting fabrication process; characterized composite material properties using SEM imaging and mechanical testing.
- Published findings in Results in Engineering (Elsevier, 2024) — first-author paper on the effect of jute stick charcoal powder on aluminum metal matrix composite characterization.

Open Channel Desktop Wind Tunnel

- Designed and built a desktop-scale open-channel wind tunnel for aerodynamic testing and demonstration.

Small-Scale Autonomous Firefighting Robot

- Built an IR-sensor-based autonomous robot for fire detection and response as part of a university robotics project.
- Additional coursework/project work: numerical analysis of raw UTM data, CAD design of mechanical parts and vehicle systems, experimental setup design for materials casting and additive manufacturing.

WORK EXPERIENCE

Data Development Lead *2025*
Takeoff Moving Company, Canada (Remote)

- Analysed fleet and operational data to support planning and decisions.
- Developed and maintained the company website and data workflows.

Executive, Operations & Supply Chain Management *May 2024 – Nov 2024*
Esquire Group Limited, Dhaka, Bangladesh

- Oversaw operations and maintenance across three business units and four factories.

- Optimized production workflows, increasing efficiency by 22%.
- Managed regular material testing and production.

Assistant Engineer *Apr 2023 – Apr 2024*
PEAL Engineering & Construction Ltd., Dhaka, Bangladesh

- Managed project execution for Horizontal Directional Drilling (HDD) projects.
- Prepared and submitted EPC tender documentation, including technical research and compliance review.
- Led procurement sourcing from local and international suppliers.

Industrial Attaché *Jan 2022 – Feb 2022*
Central Locomotive Workshop, Railway Ministry of Bangladesh

- Rotated across departments to study repair and maintenance workflows; compiled data for operational analysis.

Engineering Intern *Sep 2021 – Dec 2021*
Battery Low Interactive Ltd., Dhaka, Bangladesh

- Performed QA/QC testing on websites and VR applications, including a project delivered to the Bangladesh Army.

AWARDS & ACHIEVEMENTS

- Winner, Idea Contest and Hackathon — GPH Ispat Esho Robot Banai, 2019
- Selected as one of 30 global Space Leaders — NASA Space Apps Collective Cohort, 2023
- Completed Microsoft Undergraduate and Master's Engineering Virtual Experience Program
- Runner-up, Intra-University Cultural Contest (Instrumental), 2020

LEADERSHIP & EXTRACURRICULAR

- Volunteer Reviewer, Elsevier journal Results in Engineering (Jan 2024 – Present)
- President, BAUST Cultural Society (Jan 2022 – Feb 2023)
- Member, International Association of Engineers (IAEng), Mechanical Engineering & AI societies (2019 – Present)

References available upon request.